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Speech synthesis from neural decoding of spoken sentences

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Brain-Computer Interaction Course

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Introduction

What is the problem?

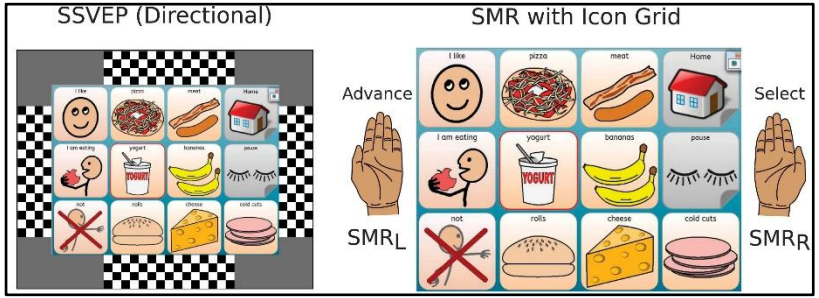
- Building a neural speech prosthetic by translating brain signals into intelligible synthesized speech

Why is this important?

- Existing methods controlled cursors to select letters or vocabulary. This work aimed at naturally decoding the user's sentences, which enables high accuracy and high rate of words transmission

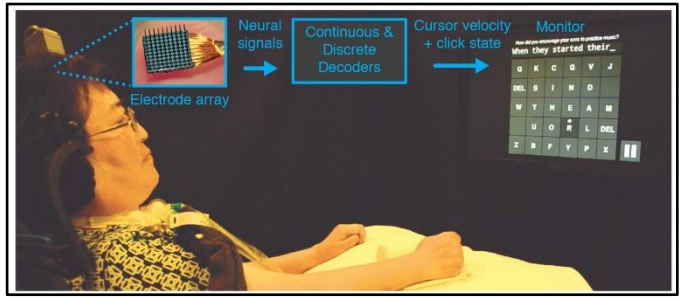
Existing related works

InvoTech (2016)



Vaughan *et al*. (2006)

Blankertz *et al*. (2006)



Pandarinath *et al*. (2017)

Introduction

What have the authors done differently to solve this problem?

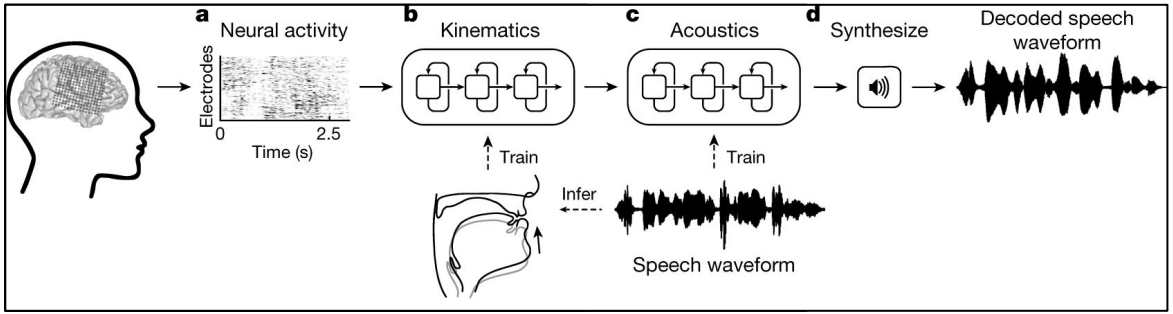
- Recorded high-density electrocorticography (ECoG) signals from participants who underwent intracranial monitoring for epilepsy treatment as they spoke sentences aloud
- Designed a recurrent neural network (RNN) that decodes cortical signals with a representation of the articulatory dynamics to synthesize audible speech

Results

Demonstrated speech synthesis using high-density, cortical recordings from the human speech cortex

- User can form sentences at normal speaking rate
- Pitch intonation as well as other natural speech patterns can be incorporated
- Possible to transfer mapping of kinematics model across participants, including the paralyzed

Speech Decoder Design



- Bidirectional long short-term memory (bLSTM) RNN decodes articulatory kinematic features from continuous neural activity recorded from vSMC, STG, and IFG
- The second bLSTM decodes acoustic features from the decoded articulatory features
- The audio signal is then synthesized from the decoded acoustic features

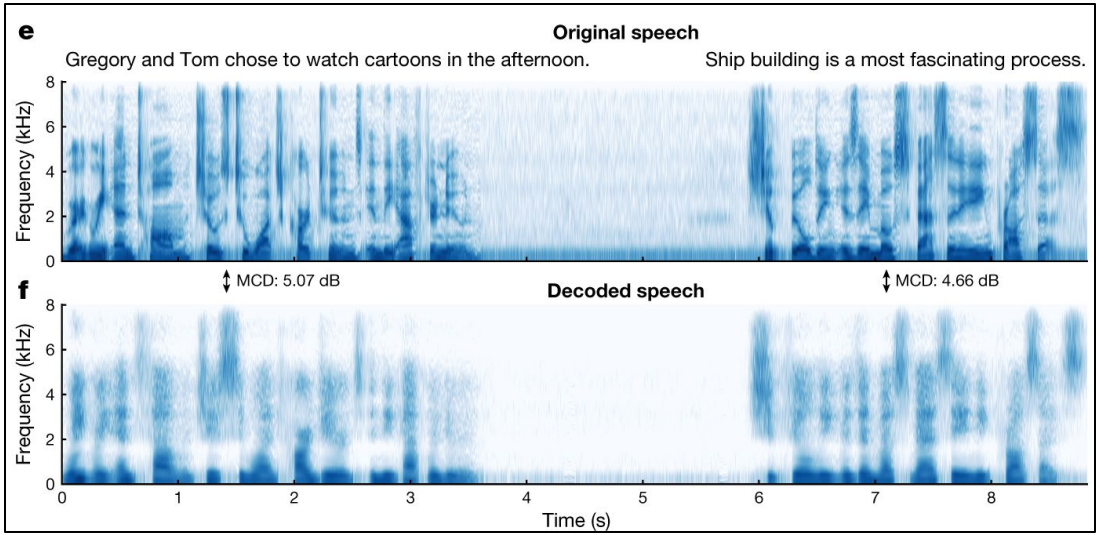
❖ LSTM: RNN with feedback connections. Can process not only single data points but also sequences of data

❖ Ventral sensorimotor cortex (vSMC): covers the primary sensory and motor areas

❖ Superior temporal gyrus (STG): responsible for processing sounds

❖ Inferior frontal gyrus (IFG): involved in language processing and speech production

Synthesis Performance



- Using the Mel log spectral approximation (MLSA), synthesized speech waveforms are constructed from acoustic features from the neural decoder
- Overall, successful reconstruction of spoken sentences through neural decoding
  - Spectrograms of the original and decoded speech show the retention of energy patterns and pauses in speech
- Mel-cepstral distortion (MCD)
  - Measure of synthesized speech quality when compared to original audio (ground truth)

$$\mathrm{MCD} = \frac{10}{\ln(10)} \sqrt{\sum_{0<d<25} (\mathrm{mc}_d^y - \mathrm{mc}_d^{\hat{y}})^2}$$

Synthesis Performance

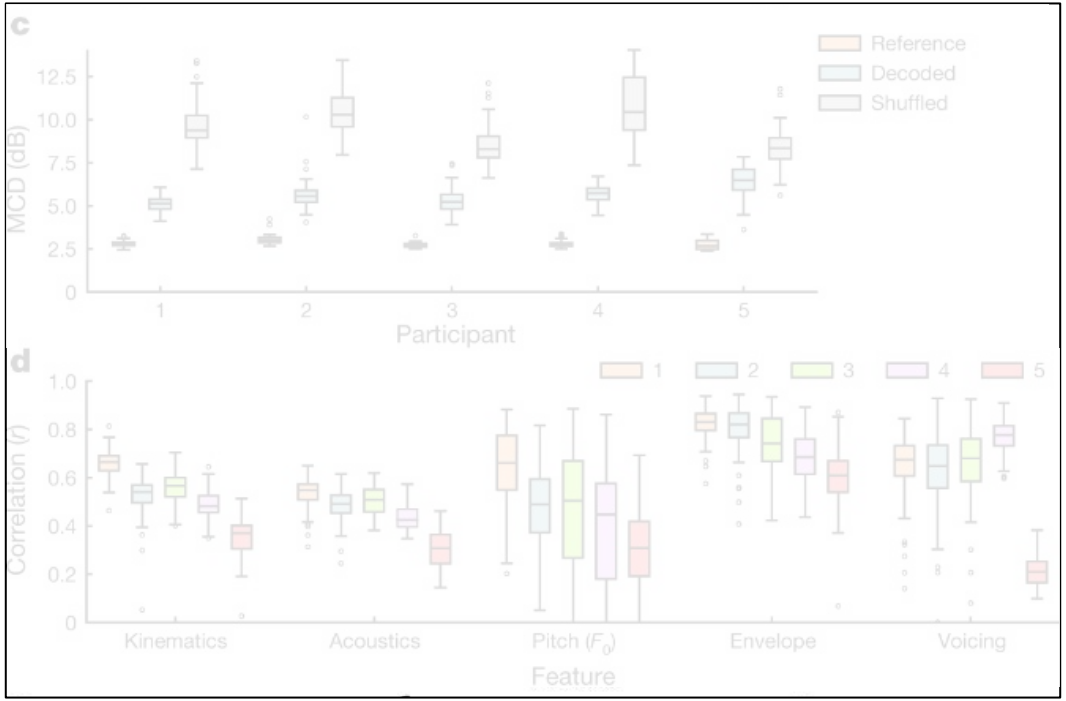
Two tasks to determine perceptual intelligibility to native English speakers

1. **Single-word identification**
   - 325 synthesized words- either 1,2,3, or 4 syllables
   - Word pool of 10, 25, or 50 words
   - As syllable length increased and word pool choices decreased, word identification performance increased
2. **Sentence-level transcription**
   - Closed looped vocabulary derived MOCHA-TIMIT dataset
   - 101 synthesized sentences. Word pool of 25 or 50 words
   - 31% median WER for 25-word pool and 53% median WER with 50-word pool
   - Smaller word pool size results in smaller WER

$$\mathrm{WER} = \frac{I + D + S}{N}$$

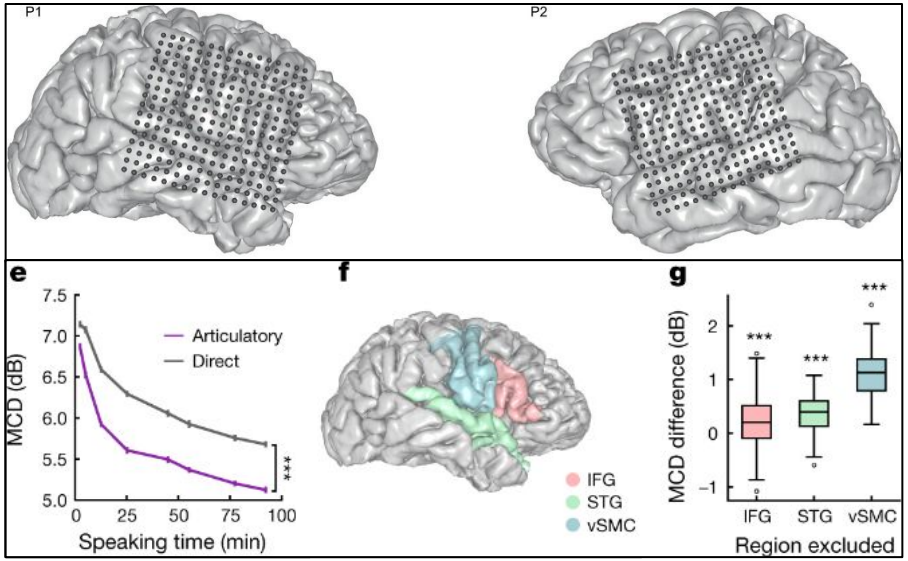
I- # of word insertions
D- # of word deletions
S- # of word substitutions for a reference sentence with *N* words

Synthesis Performance



- Spectral distortion (MCD) was measured between audio from original spoken sentences and (1) audio from articulatory kinematics, (2) audio from neural decoding, (3) audio from shuffling electrode inputs into neural decoder
- Participant 1 consistently showed highest mean correlation of original and decoded features

Decoder Characteristics



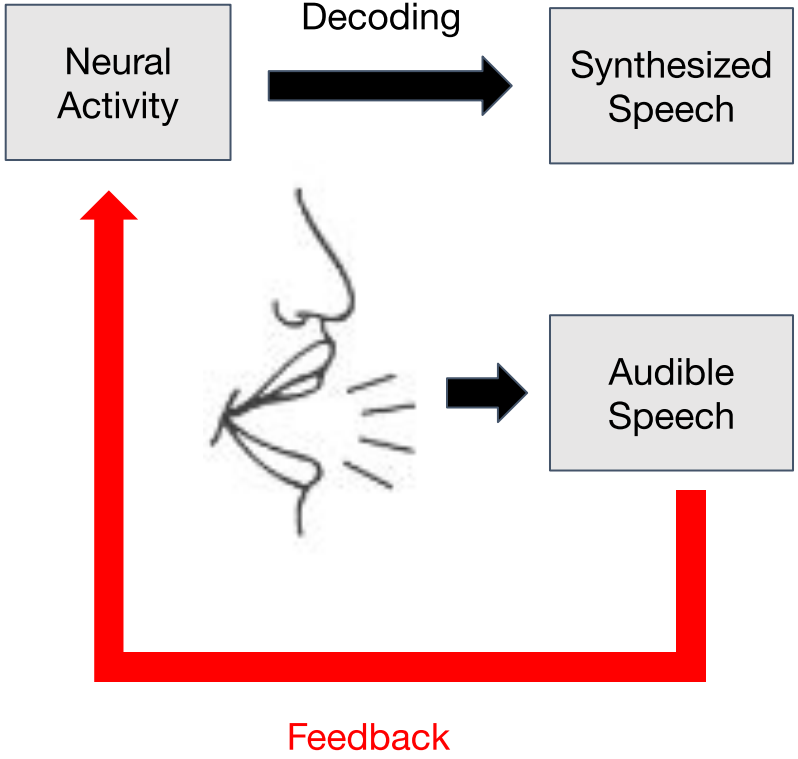
- Amount of data necessary to achieve high level of performance

➔ Found robust performance could be achieved with 25 minutes of speech, but performance continued to improve with the addition of data.

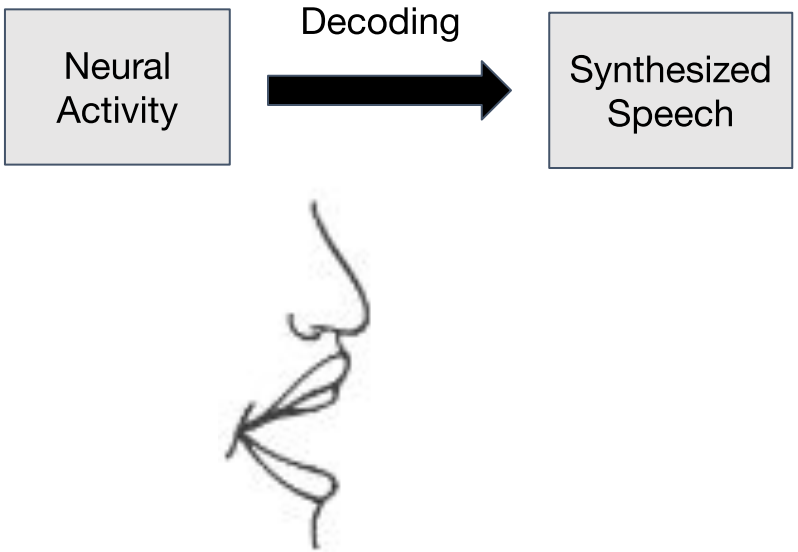
- Contribution of several anatomical regions (vSMC, STG, and IFG)

➔ Excluding the vSMC resulted in the largest decrease in performance

Synthesizing Mimed Speech



Synthesizing Mimed Speech



Synthesizing Mimed Speech

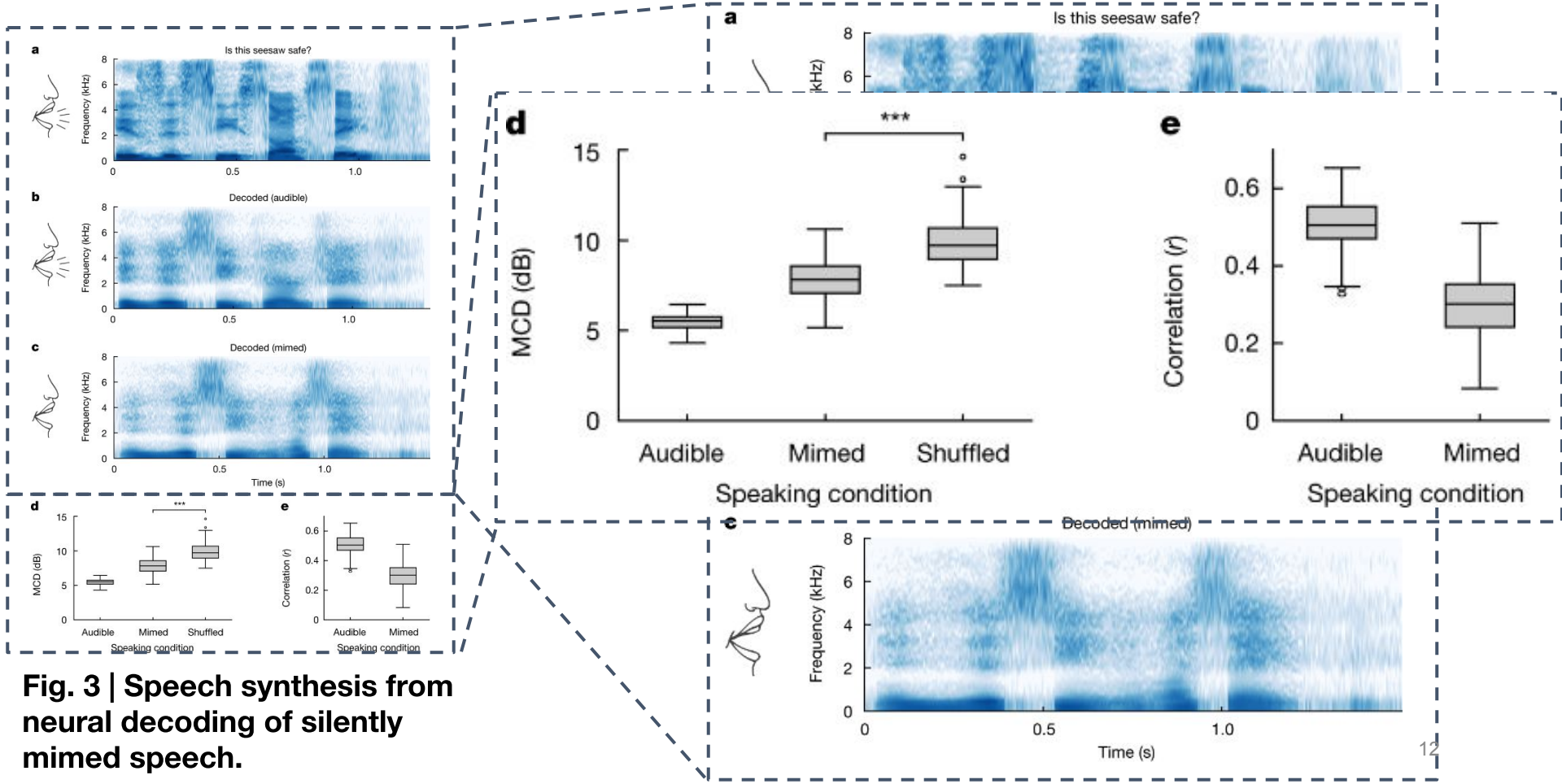


Fig. 3 | Speech synthesis from neural decoding of silently mimed speech.

State-Space of Decoded Speech Articulation

- Single principle component trajectories are able to display some high level behavioral speech behavior

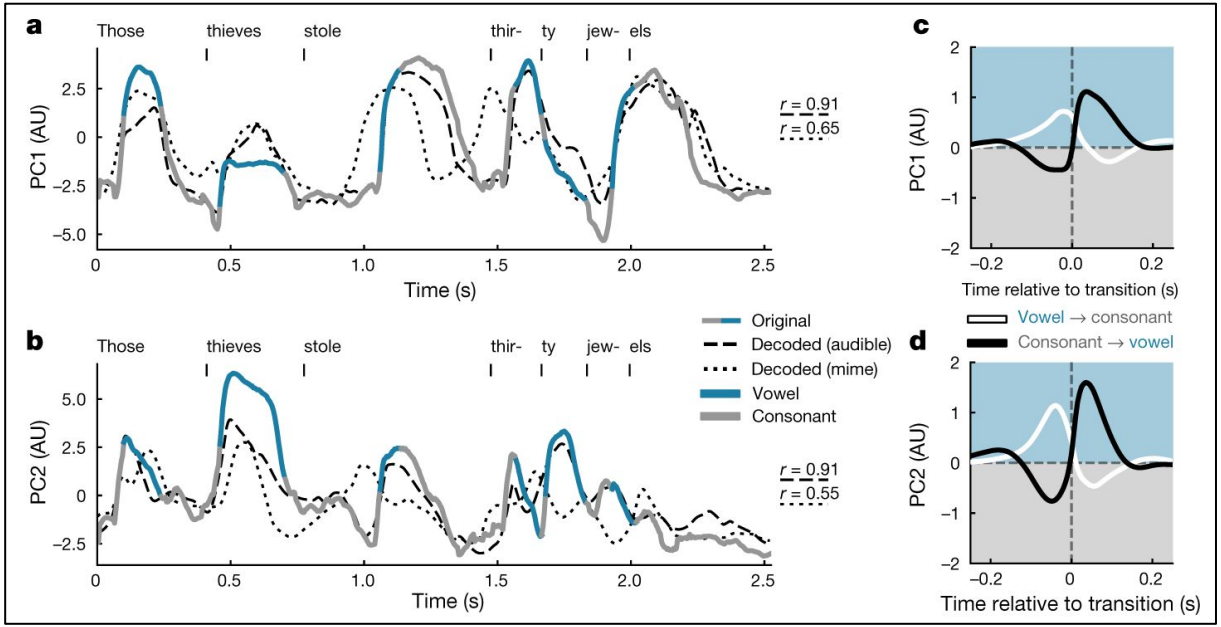


Fig. 4 | Kinematic state–space representation of speech production.

State-Space of Decoded Speech Articulation

- The kinematic trajectories were well correlated between original and decoded speech
- The decoder may be transferable between subjects.
  - Trajectories were well correlated between participants (**Figure 4, f**).
  - Transfering decoders resulted in acceptable performance (**Figure 4, g**).

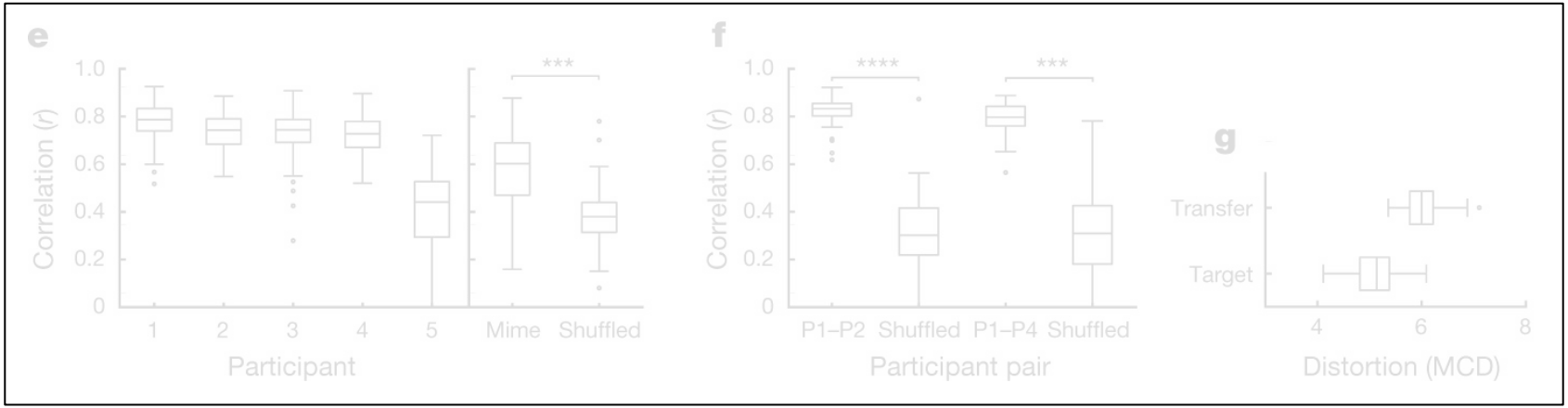


Fig. 4 | Kinematic state–space representation of speech production.

Conclusion

Demonstrated speech synthesis using high-density, cortical recordings from the human speech cortex

Advantages of the proposed speech synthesis

- User can form sentences at normal speaking rate
- Pitch intonation as well as other natural speech patterns can be incorporated
- Possible to transfer mapping of kinematics model across participants, perhaps for those who are paralyzed

Limitations

- Articulatory movement is required to train the model
- Whether the model could synthesize speech without the articulatory movement is debatable
- Some participants showed lower speech synthesis performance

References

1. Anumanchipalli, G.K., Chartier, J. & Chang, E.F. Speech synthesis from neural decoding of spoken sentences. *Nature* 568, 493–498 https://doi.org/10.1038/s41586-019-1119-1 (2019).
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4. Pandarinath, C. et al. High performance communication by people with paralysis using an intracortical brain–computer interface. *eLife* 6, e18554 (2017).

Questions

1. This is an issue with our paper too, but the physical movement of speaking is the main cause of the neural activity. For actual paralyzed / locked-in patients, do you think this approach can still be used?
2. Why didn’t they provide the accuracy of words (just phonemes)? We see high transfer rate but accuracy could be terrible.
3. Electrode placement different per person - why and how to optimize for real paralyzed people (i.e., where to put on a paralyzed person before training)?
4. In the paper, they mentioned that they excluded electrodes to exclude any auditory response. They excluded the electrodes that did not contain significant features or activity changes. Couldn’t auditory stimuli (i.e., hearing themselves speak) have elicited neural responses? I'm not sure that excluding the electrodes in this manner achieved the desired goal.
5. They say they have to manual select the onset of phonemes. How do you think this model would be translated to a real BCI?